NAD+ (Nicotinamide Adenine Dinucleotide)

What It Is

NAD+ (nicotinamide adenine dinucleotide) is a molecule found in every cell of your body that plays a central role in energy production, DNA repair, and cellular health. It is not a peptide. It is a coenzyme, which means it helps enzymes do their jobs. Without adequate NAD+, your cells cannot produce energy efficiently, repair damaged DNA, or maintain the hundreds of metabolic reactions that keep you functioning.

Most people hear about NAD+ as an anti-aging supplement, but before you decide whether it is for you, you need to understand what it actually does at the cellular level, what happens when it declines, and the different ways to address that decline.

Every single thing your body does requires energy, and that energy comes from ATP. Your cells use ATP to contract muscles, fire neurons, repair tissue, and keep you alive. Your body stores almost zero ATP. You only have about 100 grams on hand at any given time, which is basically enough energy for just a few seconds of intense activity. So your body has to constantly remake ATP all day, every day. A moderately active person cycles through their entire body weight in ATP daily.

NAD+ is essential for that ATP production process. It is a critical cofactor in the electron transport chain, which is where about 90% of your ATP gets made. When NAD+ levels drop, your energy production slows down, and that affects everything downstream.

The problem is that NAD+ levels decline significantly with age. By the time you reach your 40s, your NAD+ levels may have dropped by 50% or more compared to when you were young. This decline is associated with reduced energy, impaired cellular repair, metabolic dysfunction, and many of the hallmarks of aging.

NAD+ can be raised through oral precursor supplements (niacin, NMN, NR), through subcutaneous or intramuscular injections, or through lifestyle interventions like exercise. Each method has different absorption characteristics and practical considerations.

How It Works

NAD+ exists in two forms that cycle back and forth constantly. NAD+ is the oxidized form. During metabolism, NAD+ picks up electrons and becomes NADH. Think of NAD+ as the empty shuttle bus and NADH as the shuttle bus carrying passengers (electrons) to the power plant.

The Energy Production Pathway

Your body makes ATP through three pathways, and NAD+ is involved in all of them.

Glycolysis breaks down glucose outside of your mitochondria. It is fast, happens in milliseconds, and produces a small amount of ATP plus some NADH. This is what fuels the first 10 seconds or so of very intense activity like heavy lifting or sprinting. It cannot be sustained for long because it produces lactic acid as a waste product.

The Krebs cycle happens inside your mitochondria. It takes what glycolysis produced and fully oxidizes it. This is where most of your NADH gets made. NADH is the carrier molecule that picks up electrons and transports them to the next stage.

The electron transport chain happens on the inner membrane of your mitochondria. It uses NADH to pump protons and create ATP. This is where about 90% of your ATP comes from. When NADH gives up its electrons at the end of this process, it becomes NAD+ again. And this cycle repeats thousands of times per day in every cell.

In healthy cells, you have about 700 parts NAD+ for every one part NADH. This ratio matters. If you have an abundance of NAD+, your cells produce energy very efficiently. If NAD+ is scarce, energy production becomes limited and cellular stress increases. This is the difference between having endurance, fast recovery, and mental clarity versus feeling fatigued, foggy, and slow to recover.

Why NAD+ Declines

Three things cause the decline. First, certain enzymes in your body (particularly CD38 and PARPs) consume more NAD+ as you age because the processes they support, like DNA repair and immune function, become less efficient and create more demand. CD38 levels increase with age and chronic inflammation, accelerating NAD+ depletion.

Second, there is an enzyme called NAMPT that makes NAD+ through the salvage pathway, which is responsible for about 80% of your NAD+ production. NAMPT becomes less efficient as you age.

Third, when your mitochondria get damaged, they cannot recycle NAD+ as effectively. This creates a vicious cycle because lower NAD+ means less efficient energy production, which means more mitochondrial stress, which means even lower NAD+.

Sirtuin Activation

Beyond energy production, NAD+ activates a family of proteins called sirtuins (SIRT1 through SIRT7) that regulate cellular stress responses, metabolism, and longevity. Sirtuins require NAD+ as a co-substrate to function. When NAD+ levels are low, sirtuin activity decreases, which impairs your body's ability to respond to stress, repair damage, and maintain metabolic health.

Sirtuins are involved in DNA repair and genomic stability, inflammation control, mitochondrial biogenesis (making new mitochondria), fat metabolism and insulin sensitivity, and circadian rhythm regulation.

DNA Repair

NAD+ is consumed by PARP enzymes (poly ADP-ribose polymerases) when they repair DNA damage. As DNA damage accumulates with age and NAD+ levels decline, the repair machinery becomes less effective. This creates another vicious cycle where damaged DNA goes unrepaired, contributing to cellular dysfunction and aging.

Benefits

Increased Cellular Energy

By supporting mitochondrial function, NAD+ supplementation may improve energy levels and reduce fatigue. Many people report feeling more energetic and mentally sharp after raising their NAD+ levels.

Improved Metabolic Health

Preclinical and early human studies suggest NAD+ boosting can improve insulin sensitivity and support healthy lipid metabolism. A study in overweight women with prediabetes showed improved insulin sensitivity in skeletal muscle after 10 weeks of NMN supplementation at 250 mg daily.

Enhanced Cognitive Function

NAD+ supports brain health through multiple mechanisms including energy production, DNA repair, and sirtuin activation in neurons. Users of NAD+ therapy often report improved mental clarity and reduced brain fog.

DNA Repair and Cellular Maintenance

By providing fuel for DNA repair enzymes, adequate NAD+ helps maintain genomic integrity and supports healthy cellular function over time.

Cardiovascular Support

NAD+ plays important roles in heart health. Preclinical research shows NAD+ replenishment can protect against cardiovascular disease and support healthy blood pressure.

Recovery Support

NAD+ therapy has been used in clinical settings to support recovery from substance dependence, helping reduce withdrawal symptoms and support brain health during detoxification. This should only be done under medical supervision.

What to Expect

First Week

Increased energy is commonly reported, especially with injectable routes. Some people notice improved sleep quality and mental sharpness. Effects can be subtle or pronounced depending on your baseline NAD+ status. Those with significantly depleted levels tend to notice more dramatic improvements.

Weeks 2 to 4

Sustained energy improvements. Better exercise recovery is often noted. Cognitive benefits may become more noticeable as cellular repair processes ramp up.

Long Term

The cellular level benefits (DNA repair, mitochondrial support, sirtuin activation) accumulate over time but are not directly perceptible. Continued use maintains elevated NAD+ levels and their associated benefits.

After Stopping

NAD+ levels gradually return to baseline. There is no withdrawal or rebound effect, but benefits will diminish over weeks to months without continued supplementation.

The Catch

If your mitochondria are damaged, just adding more NAD+ might not work. Converting NAD+ precursors into usable NAD+ requires ATP. If your mitochondria cannot produce ATP efficiently, they cannot use NAD+ precursors effectively. This is exactly why some people supplement with NAD+ and do not feel anything. The machinery is too broken to use the fuel. This is where SS-31 comes in first for the repair phase.

What the Science Shows

NAD+ research is evolving rapidly. Here is where the evidence currently stands.

NAD+ Precursors Raise Blood NAD+ Levels (Human Studies)

Multiple human trials have confirmed that oral NMN and NR supplementation successfully raises blood NAD+ levels. Studies show increases of 40 to 90% with consistent supplementation. However, the degree of increase in tissues (where it actually matters) is less clear.

Metabolic Benefits (Human Trial, 2021)

Yoshino and colleagues published in Science a study of overweight or obese women with prediabetes showing that 10 weeks of NMN supplementation (250 mg daily) improved insulin sensitivity in skeletal muscle. This is one of the stronger pieces of human evidence for a clinically meaningful endpoint.

NMN Meta-Analysis (2024)

A systematic review with meta-analysis of randomized controlled trials published in Critical Reviews in Food Science and Nutrition found that NMN supplementation significantly elevated blood NAD levels. However, most clinically relevant outcomes (body composition, physical performance, metabolic markers) were not significantly different between NMN supplementation and control groups. This is an important reality check: raising NAD+ levels does not automatically translate to measurable clinical improvements.

The Gut Conversion Problem

Recent research shows that most oral NMN and NR supplements get converted to nicotinic acid (niacin) by gut bacteria before they can be absorbed intact. A study published in Science Advances showed that NR and NMN facilitate NAD+ synthesis via enterohepatic circulation, with gut bacteria deamidating these compounds to nicotinic acid during absorption. So you may be paying significantly more for compounds that your gut is turning into niacin anyway.

Liposomal NMN formulations may partially bypass this problem. A 2025 study in healthy men over 40 showed that liposomal NMN significantly increased NAD+ compared to non-liposomal NMN and placebo.

Niacin: The Cheapest Proven Option

Niacin (vitamin B3) feeds directly into the Preiss-Handler pathway and gets converted into NAD+. It is a proven, cost-effective way to raise NAD+ levels. The catch is the niacin flush, which is red, hot, itchy skin for about 20 to 30 minutes after you take it. Most people cannot tolerate that. If you can handle the flush, niacin is your cheapest and most proven option for raising NAD+ levels.

Direct NAD+ Injection

When you inject NAD+ directly, you are not feeding into any of the biosynthesis pathways. You are not waiting for your body to convert anything. You are putting NAD+ straight into your bloodstream. This bypasses the gut bacteria problem, avoids the flush, and gets NAD+ directly where it needs to go. While clinical experience suggests benefits, rigorous controlled trials specifically on subcutaneous NAD+ are limited.

NAMPT Suppression Risk

This is critical. Chronic NAD+ precursor supplementation can suppress your body's natural production of NAMPT, which is the key enzyme in the salvage pathway that makes about 80% of your NAD+. A human study found that NR supplementation decreased NAMPT levels despite increasing NAD+ metabolites. Long-term supplementation could actually make your body worse at producing NAD+ naturally. If you stop supplementing after chronic use, you may be in a worse position than before you started.

The solution is to cycle your supplementation. Do 8 to 12 weeks on, then 4 to 8 weeks off. Exercise actually increases NAMPT expression by over 100% and raises muscle NAD+ by over 100%. Training, nutrition, and sleep always remain the primary focus.

Animal Studies

In animal studies, NAD+ supplementation extended lifespan and healthspan, improved physical function, and reversed some aspects of age-related decline. Whether these dramatic effects translate to humans is still being studied.

What People Report

The following is aggregated from external platforms including Reddit, peptide forums, and community conversations. This is anecdotal data and does not carry the same weight as published research.

Energy and Mental Clarity

The most commonly reported benefit across all administration routes is improved energy and mental clarity. Injectable users tend to report faster and more noticeable effects compared to oral precursors.

The Injection Sting

Subcutaneous NAD+ injections are often described as more uncomfortable than typical peptide injections, with a burning or stinging sensation. Injecting slowly, using a slightly larger needle gauge, and warming the solution to room temperature before injection are commonly recommended strategies.

Variable Responses to Oral Precursors

Reports on oral NMN and NR are mixed. Some users report clear improvements in energy and recovery, while others notice nothing. This is consistent with the scientific finding that oral precursors may largely be converted to niacin in the gut and that individual variation in gut microbiome composition affects conversion efficiency.

Sleep Effects

Some users report improved sleep quality, while others report difficulty sleeping if they take NAD+ later in the day. Morning administration is generally recommended.

Niacin Flush

Users who choose niacin as their NAD+ precursor report the flush diminishes over time with consistent daily use. Starting with lower doses (100 to 250 mg) and gradually increasing helps build tolerance.

Dosing Protocol

Understanding the Different Approaches

Unlike some peptides that only work when dysfunction is present, NAD+ decline affects everyone with age. By your 40s, NAD+ levels may have dropped 50% or more from youthful levels. This means most adults over 40 can potentially benefit from NAD+ support regardless of whether they have symptoms.

Subcutaneous Injection

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LOADING PROTOCOL
Dose:           100 to 200 mg daily
Duration:       7 to 10 consecutive days
Purpose:        Rapidly restore depleted cellular reserves
MAINTENANCE PROTOCOL
Dose:           50 to 100 mg
Frequency:      1 to 3 times per week
Timing:         Morning preferred (may interfere with sleep if taken late)
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Intramuscular Injection

50 to 100 mg, 1 to 3 times per week. Similar dosing to subcutaneous but absorbed slightly faster.

Oral Precursors

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NIACIN (Vitamin B3)
Starting dose:  100 to 250 mg daily (build tolerance for flush)
Standard dose:  500 to 1000 mg daily
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Note:           Cheapest and most proven oral option
Limitation:     Niacin flush (redness, warmth, itching for 20-30 min)
NMN (Nicotinamide Mononucleotide)
Starting dose:  250 to 500 mg daily
Standard dose:  500 to 1000 mg daily
Note:           More expensive, may largely convert to niacin in gut
Limitation:     NMN regulatory status uncertain (FDA flagged in 2022)
NR (Nicotinamide Riboside)
Starting dose:  250 to 500 mg daily
Standard dose:  500 to 1000 mg daily
Note:           Available as dietary supplement (Tru Niagen, etc.)
Limitation:     Similar gut conversion concerns as NMN
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Cycling (Non-Negotiable for Ongoing Use)

Do 8 to 12 weeks on NAD+ supplementation, then 4 to 8 weeks off. This prevents NAMPT suppression and gives your body's natural production pathways a chance to recover. Exercise during the off-cycle helps restore natural NAD+ production.

Reconstitution (Injectable)

NAD+ requires more bacteriostatic water than most peptides. It is a large molecule that can crystallize or precipitate out of solution when the concentration is too high. The standard 2 mL reconstitution used for most peptides does not work here.

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RECONSTITUTION
500 mg vial:    Add 3 mL bacteriostatic water
                Concentration: 166.7 mg/mL
                For 100 mg dose, draw 60 units on insulin syringe

1000 mg vial:   Add 6 mL bacteriostatic water
                Concentration: 166.7 mg/mL
                For 100 mg dose, draw 60 units on insulin syringe
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NAD+ typically comes in 10 mL glass vials, so both volumes fit with room to spare. Let the water run down the inside wall of the vial. Do not spray directly on the powder. Swirl gently until fully dissolved. Do not shake.

After reconstitution, refrigerate at 2 to 8 degrees Celsius. Use within 14 days (not the typical 28 days for most peptides). Do not freeze. Protect from light. Before every injection, inspect the solution. It should be clear and colorless. If you see any cloudiness, discoloration, or particles, the NAD+ has precipitated out of solution and that vial should be discarded.

Stacking Context

The Cellular Energy Protocol

This is the framework that ties all the mitochondrial compounds together. It happens in two phases.

Phase 1: Repair (Weeks 1 through 8) SS-31 to repair mitochondrial membrane structure. NAD+ (injectable or niacin) to provide substrate through the biosynthesis pathways. 5-amino-1MQ to block the drainage of NAD+ building blocks in the salvage pathway. Creatine to support the overall ATP system.

Phase 2: Optimize (Weeks 5 through 12) Replace SS-31 with MOTS-c, which helps your body optimize how efficiently the repaired mitochondria run. Continue NAD+ and creatine. Continue 5-amino-1MQ if desired.

NAD+ + SS-31

SS-31 repairs the mitochondrial structure. NAD+ provides the fuel. If your mitochondria are damaged, just adding NAD+ gives limited results because the machinery cannot use the fuel efficiently. Fix the structure first with SS-31, then support with NAD+.

NAD+ + Epithalon

This combination targets multiple aspects of cellular aging. Epithalon supports telomere maintenance while NAD+ supports mitochondrial function and DNA repair. Different mechanisms, complementary effects.

NAD+ + 5-Amino-1MQ

5-amino-1MQ inhibits NNMT, an enzyme that normally takes nicotinamide (the building block of NAD+) and converts it into waste that gets excreted. By blocking that drainage, more nicotinamide stays available for NAD+ production through the salvage pathway.

NAD+ + Creatine

Creatine helps your body produce phosphocreatine, which stores ATP around cells for rapid use. This makes you able to produce energy faster than mitochondria can for explosive power and also supports mitochondrial biogenesis. Creatine supports the ATP system that NAD+ feeds into.

NAD+ + GLP-1 Agonists

No interaction concerns. Different mechanisms entirely.

NAD+ + Growth Hormone Peptides

No interaction concerns. GH peptides require fasting. NAD+ does not. Take them on their own schedules.

Common Questions

What is the difference between NAD+, NMN, and NR? NAD+ is the active molecule your cells use. NMN and NR are precursors, meaning your body converts them into NAD+. NMN is one step away from becoming NAD+. NR converts to NMN inside your cells, then becomes NAD+. Oral NAD+ itself is poorly absorbed, which is why precursors or injectable forms are used.

Are injections better than oral supplements? Injections provide high bioavailability directly into the bloodstream and bypass the gut conversion problem. Whether this translates to greater intracellular NAD+ than oral precursors is still debated. Some experts believe oral precursors may actually be more effective at raising NAD+ inside cells because they enter cells before conversion.

Why not just take niacin? If you can tolerate the flush, niacin is actually your cheapest and most proven option. The flush diminishes over time with consistent use. The main reason people choose NMN, NR, or injectable NAD+ is to avoid that flush.

Why do I need to cycle NAD+? Chronic supplementation can suppress NAMPT, the key enzyme in the salvage pathway that produces about 80% of your NAD+. If you suppress it long enough and then stop supplementing, your body may produce less NAD+ naturally than before you started. Cycling (8 to 12 weeks on, 4 to 8 weeks off) prevents this.

Can I take NAD+ if my mitochondria are damaged? You can, but you may not get much benefit from it alone. Converting NAD+ precursors into usable NAD+ requires ATP, and damaged mitochondria cannot produce ATP efficiently. Consider using SS-31 first to repair the structural damage, then add NAD+ once the machinery can actually use it.

When should I take it? Morning is generally preferred because NAD+ boosts cellular energy production. Taking it later in the day may interfere with sleep in some individuals.

Side Effects

In Published Research

Oral NAD+ precursors (NMN, NR) have been studied at doses up to 1000 to 2000 mg daily and appear safe in the short to medium term. No significant toxicity has been reported at standard doses.

Common (usually mild):

- Flushing and warmth (primarily with niacin)
- Nausea
- Headache
- Fatigue
- Stomach discomfort
- Injection site reactions (redness, tenderness, stinging)

Important notes:

- Side effects are often dose and rate dependent
- Starting with lower doses helps assess tolerance
- Most side effects resolve quickly
- Long-term safety data beyond 12 months is still limited

What Users Report

Injectable NAD+ is commonly described as more uncomfortable to inject than other peptides. A stinging or burning sensation at the injection site is frequently reported. Injecting slowly helps. Some users report mild nausea, especially at higher doses.

Contraindications and Precautions

Do Not Use If You Have:

- Active cancer (effects on cancer cell metabolism are unclear, and cancer cells may also benefit from increased NAD+)
- Severe liver disease (liver is involved in NAD+ metabolism)
- Severe kidney disease (kidneys help clear NAD+ metabolites)
- Known hypersensitivity to NAD+ or any component

Use Caution With:

- Pregnancy or breastfeeding (no safety data)
- Bleeding disorders (if using injectable form)
- Medications that affect cellular metabolism (consult your doctor)
- Gout or hyperuricemia (niacin can raise uric acid levels)

Drug Interactions:

- Niacin may interact with blood pressure medications, blood thinners, and diabetes medications
- No well-established interactions with injectable NAD+ due to limited research
- If taking metformin, consult your doctor (both affect cellular metabolism)

Support Nutrients

Trimethylglycine (TMG) Some practitioners recommend TMG when using high-dose NAD+ precursors to support methylation pathways. NAD+ metabolism consumes methyl groups, and TMG provides additional methyl donors.

Quercetin and Apigenin Both are natural CD38 inhibitors. CD38 is the enzyme that breaks down NAD+, and its levels increase with age. Inhibiting CD38 may help maintain higher NAD+ levels.

Resveratrol Activates sirtuins alongside NAD+. Some protocols combine resveratrol with NAD+ precursors for synergistic effects, though the human evidence for resveratrol itself remains mixed.

Creatine Supports the ATP system that NAD+ feeds into. Helps store ATP around cells for rapid use and supports mitochondrial biogenesis.

Storage and Handling

Injectable NAD+ (After Reconstitution)

Refrigerate at 2 to 8 degrees Celsius. Do not freeze. Use within 14 days (shorter stability window than most peptides due to the molecule's size and tendency to precipitate). Protect from light. Inspect for clarity before each use. If cloudy, discolored, or if you see particles, the NAD+ has crashed out of solution. Discard the vial.

Lyophilized Powder (Before Reconstitution)

Store at room temperature or refrigerated. Keep dry and away from light. Stable for much longer in lyophilized form.

Oral Supplements

Store in a cool, dry place. Follow manufacturer recommendations. NMN in particular can degrade if exposed to heat or moisture.
